

concordance

August 22, 2025

1 Concordance IVFPQ vs. KNN

SCimilarity uses TileDB KNN IVF_FLAT index which provides exact vector similarity search.

Cytoverse quantizes the vectors via Product Quantization (PQ) and then performs an approximate lookup based on partitions in an Inverted File Index (IVF) fetched over HTTP.

Note: The memory calculations use `tracemalloc.test_traced_memory()` and are only valid from a fresh run of this notebook on a restarted kernel. We call `gc.collect()` to attempt a clean measurement between each set of parameters

Loaded test data shape: (100, 44942)

1.1 Compute Exact Neighbors via KNN

KNN execution time: 42.6798 seconds
KNN memory (~bandwidth) usage: 1085.57 MiB
KNN peak memory usage: 1085.57 MiB
KNN on-disk storage: 9023.19 MiB

1.2 Compute Approximate IVFPQ Neighbors

IVF partitions: 0%| | 0/3 [00:00<?, ?it/s]
Probes (n_sub=8): 0%| | 0/8 [00:00<?, ?it/s]
Probes (n_sub=16): 0%| | 0/8 [00:00<?, ?it/s]
Probes (n_sub=32): 0%| | 0/8 [00:00<?, ?it/s]

1.3 Summary

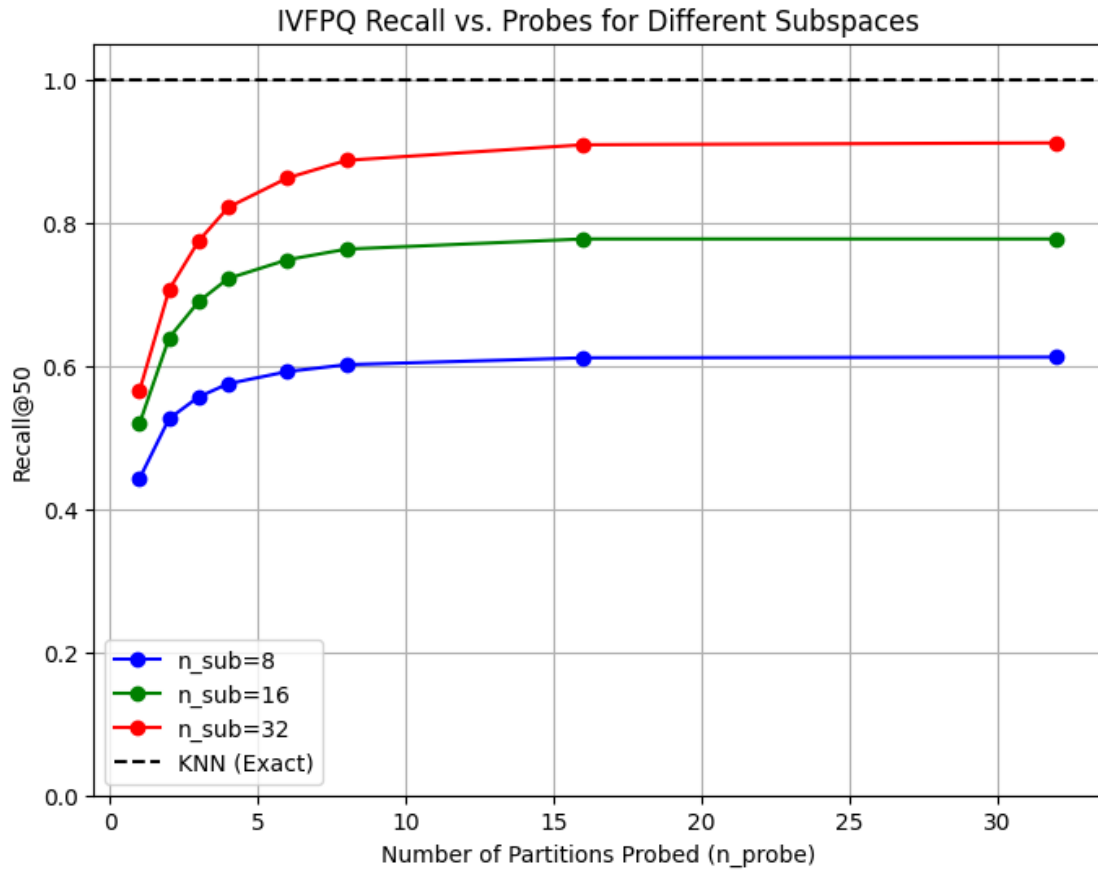
	n_sub	n_probe	Recall@K	Query Time (s)	Speedup	Storage (MB)	\
0	8	1	0.442	1.84	23.20	92.30	
1	16	1	0.519	1.91	22.32	152.71	
2	32	1	0.565	1.93	22.16	273.54	
3	KNN	N/A	1.000	42.68	1.00	9023.19	

	Storage Reduction	Bandwidth (MB)
0	97.76	3.05
1	59.09	3.05

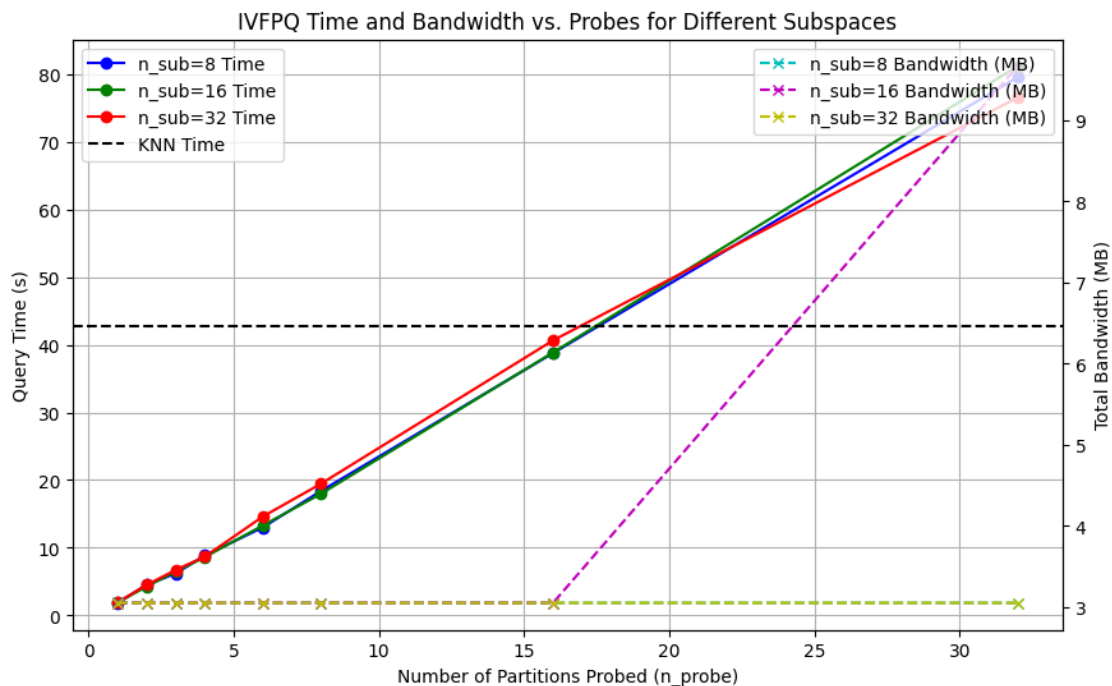
2	32.99	3.05
3	1.00	0.00

1.4 Recall vs. Probes

IVFPQ recall improves with more probes but plateaus. Fewer subspaces trade accuracy for compression reducing server storage and bandwidth.

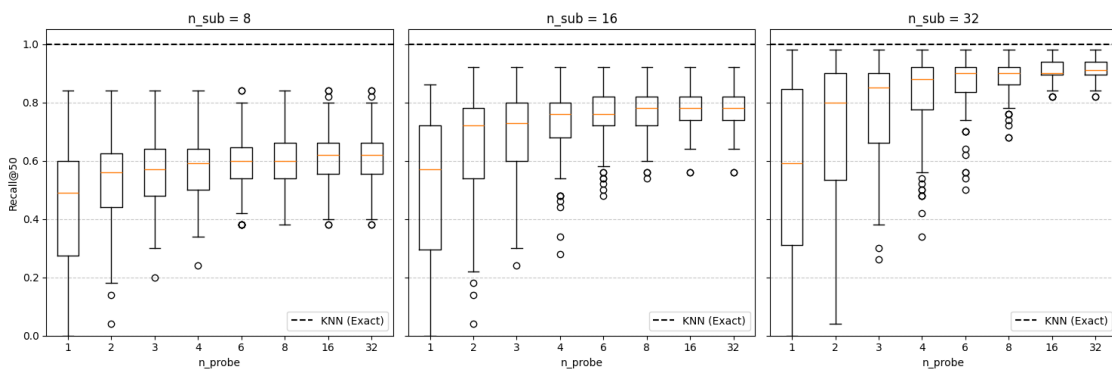


1.5 Time and Bandwidth vs. Probes



1.6 Recall@K by IVGPQ Parameter

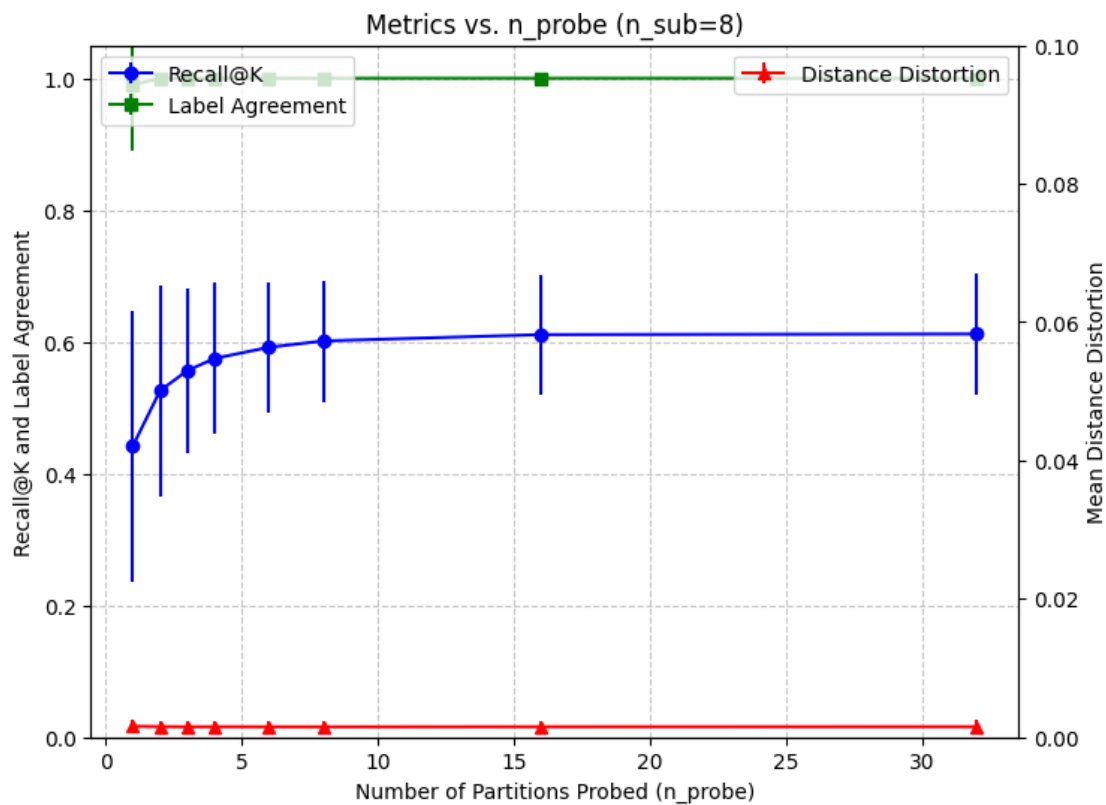
Distribution of Recall@K across all queries for each n_probe value grouped by n_sub . Shows variability in recall (not just the mean) across queries, helping researchers assess consistency for single-cell RNA-seq labeling in a web app vs. local KNN.

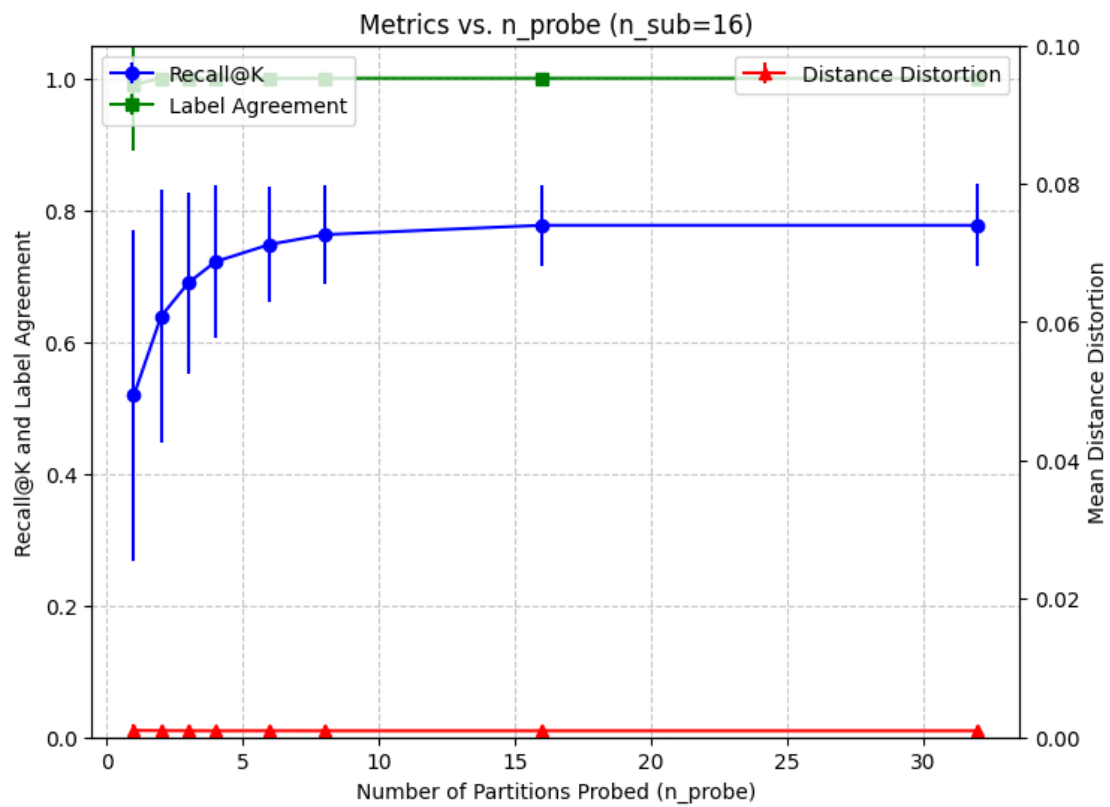


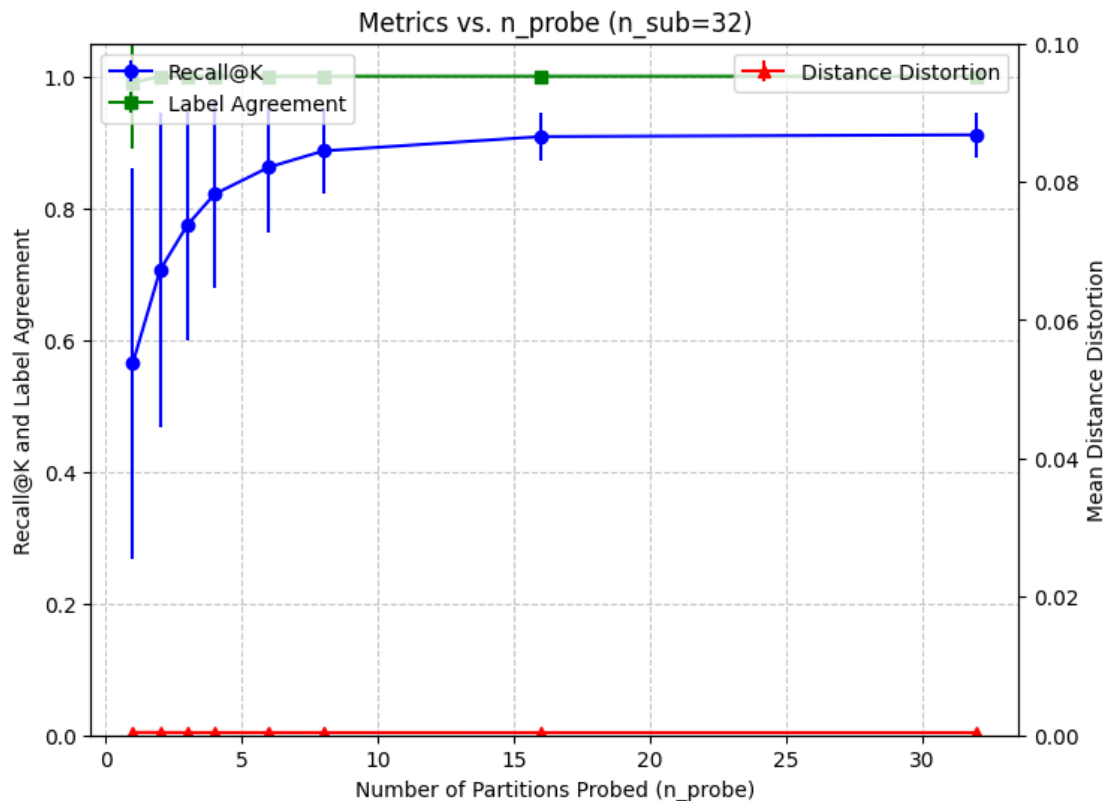
1.7 Mean Metrics vs. IVFPQ Parameter

Visualizes how mean Recall@K, Label Agreement, and mean Distance Distortion vary with n_probe for each n_sub , with error bars showing standard deviation across queries. This helps researchers

evaluate tradeoffs in accuracy (recall and label agreement) and distortion for IVFPQ in a web app context for single-cell RNA-seq labeling, compared to a local KNN implementation. The plot uses a twin-axis format: Recall@K and Label Agreement on the left (0-1 scale), Distance Distortion on the right (data-driven scale).







1.8 Distance Distortion vs. Recall@K

Relationship between mean distance distortion (for matched neighbors) and Recall@K for each query, with points colored by the average nearest neighbor distance in single-cell RNA-seq data (nn_dist_SC, reflecting query “difficulty”). This helps researchers understand if queries with farther neighbors (harder cases) exhibit higher distortion or lower recall in IVFPQ compared to KNN, relevant for web app vs. local tool tradeoffs.

